



Demonstration of Proposed Features

Date	5 July 2026
Team ID	XXXXXX
Project Name	AI Learning Support Assistant
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Emotion Prediction using BiLSTM & BERT	Predicts the user's learning emotion using BiLSTM and BERT models with confidence scores.	Implemented	Yes	Successfully demonstrated with multiple user inputs.
2	Personalized Learning Guidance	Generates learning assistance using Google Gemini API based on the detected emotion and learning problem.	Implemented	Yes	AI-generated responses displayed correctly.
3	Mixed Emotion Detection	Displays primary and secondary emotions with confidence values.	Implemented	Yes	Successfully tested using different learning scenarios.
4	Interaction History	Stores previous learning interactions and allows users to reopen them for review.	Implemented	Yes	CSV-based history loading and viewing demonstrated successfully.

5	Analytics Dashboard	Displays emotion distribution, confidence trends, subject analytics, and summary statistics using Plotly.	Implemented	Yes	Interactive charts and summary metrics verified.
6	Responsive Streamlit User Interface	Provides an intuitive interface with sidebar navigation, analytics, and mobile-friendly layout.	Implemented	Yes	Verified on desktop and mobile browsers during Epic 6 testing.

Feature Implementation Summary:

Metric	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

Demonstration Evidence

The following features were demonstrated during the final project validation:

- ✓ Subject selection and learning problem input
- ✓ BiLSTM emotion prediction
- ✓ BERT emotion prediction
- ✓ Mixed emotion detection
- ✓ Google Gemini AI learning guidance
- ✓ Emotion confidence visualization
- ✓ Plotly-based analytics dashboard
- ✓ Previous interaction history viewer
- ✓ CSV download functionality
- ✓ Clear history functionality
- ✓ Model caching and performance optimization
- ✓ Responsive UI on desktop and mobile devices
- ✓ End-to-end workflow validation (Epic 6)

Overall Result

All proposed functional features for the **AI Learning Support Assistant** were successfully implemented, tested, and demonstrated. The application met the planned project objectives, with all major modules functioning correctly during the final demonstration and validation.